Ör

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx \qquad \begin{vmatrix} \cos x = t \\ -\sin x\,dx = dt \end{vmatrix}$$

$$= \int \frac{-dt}{t} = -\int \frac{dt}{t} = -\ln|t|$$

$$= -\ln|\cos x| + c$$

Ör

$$\int \sec^3 x\,dx = \int \sec^2 x \cdot \sec x\,dx$$

u $\quad dv$

yada

$$\ln|\sec x| + c$$

$u = \sec x$

$du =$



Ör

$$\int \frac{dx}{\sqrt{1-x^2}}$$

$x = \sin t$

$dx = \cos t\,dt$

$\sqrt{1-x^2} = \sqrt{1-\sin^2} = \cos x$

$$= \int \frac{\cos t\,dt}{\cos t} = \int dt = t + c$$

$x = \sin t \Rightarrow t = \arcsin x$

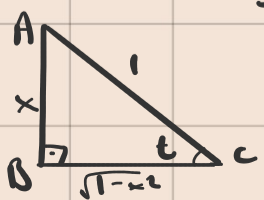
Ör

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \int \frac{a \cdot \cos t\,dt}{a \cdot \cos t} = \int dt = t + c$$

Ör

$$\int \sqrt{1-x^2}\,dx = \int \cos t \cdot \cos t\,dt = \int \cos^2 t\,dt = \int \frac{1+2\cos^2 x}{2}\,dt$$

$$= \frac{1}{2}\int dt + \frac{1}{2}\int \cos 2t\,dt = \frac{1}{2}t + \frac{1}{2}\cdot\frac{1}{2}\cdot \sin 2t + c = \frac{1}{2}\arcsin x + \frac{1}{2}\cdot x\cdot\sqrt{1-x^2} + c$$

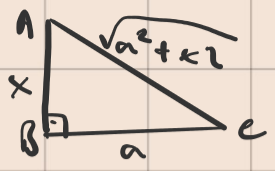


örn

$$\int \frac{dx}{\sqrt{1+x^2}} = \int \frac{\sec^2 t\, dt}{\sec t} = \int \sec t\, dt = \ln|\sec t + \tan t| + c$$

$x = \tan t$, $\sqrt{1+x^2} = \sec t$

$dx = \sec^2 t\, dt$



$x = a\tan t$

$a^2 + x^2 = a^2 + a^2\tan^2 t$

$= a^2(1+\tan^2 t)$

$= a^2 \sec^2 t$

$\sqrt{a^2+x^2} = a\sec t$

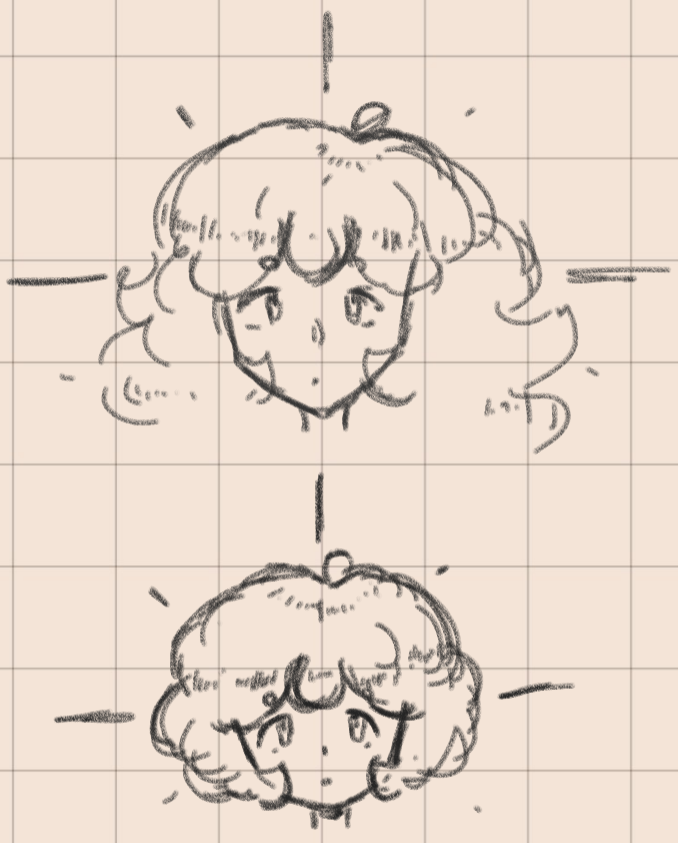
$x = a\tan t$, $dx = a\sec^2 t\, dt$

$\sqrt{a^2-u^2}$, $u = a\sin t$

$\sqrt{a^2+u^2}$, $u = a\tan t$

$\sqrt{u^2-a^2}$, $u = a\sec t$

dönüşüm yap



Reimann Toplamı ve Belirli İntegral

$y = f(x)$, $[a,b]$ aralığında sınırlı bir fonksiyon olsun.

($f(x) \to \infty$ olmayacak)

$a = x_0 < x_1 < \dots < x_n = b$ ile $p = \{x_0, \dots, x_n\}$ bölüntüsünü oluşturalım.

$[x_0, x_1], [x_1, x_2] \cdots [x_{n-1}, x_n]$

$x_0 \le z_1 \le x_1$, $x_1 \le z_2 \le x_2$, $\dots$ $z_i \in [x_{i-1}, x_i]$

$$\sum_{i=1}^{n} f(z_i)\cdot \Delta x_i = f(z_1)\cdot\Delta x_1 + f(z_2)\cdot\Delta x_2 + \cdots + f(z_n)\cdot\Delta x_n$$

(Δx_i, i. eşit uzunluğu) Bu toplama, $f(x)$, $[a,b]$ aralığındaki Riemann toplamı denir.

Belirli İntegral

Örn
$$\int_{\pi/2}^{2\pi/3} \frac{dx}{1+\sin x-\cos x} = \left(\ln\left|\frac{\tan\frac{x}{2}}{1+\tan\frac{x}{2}}\right|\right)\Bigg|_{\pi/2}^{2\pi/3} = \ln\frac{\sqrt{3}}{1+\sqrt{3}} + \ln 2$$

$t = \tan\frac{x}{2}$

$dx = \frac{2dt}{1+t^2} = \frac{dt}{t(t+1)}$

Örn
$$\int_0^1 \sqrt{1-x^2}\,dx = \int_0^1 \sqrt{\cos^2 t}\cdot\cos t\,dt = \int_0^1 (\cos t)^2 = \int_0^1 \frac{1+\cos 2t}{2}dt$$

$x = \sin t$

$dx = \cos t\,dt$



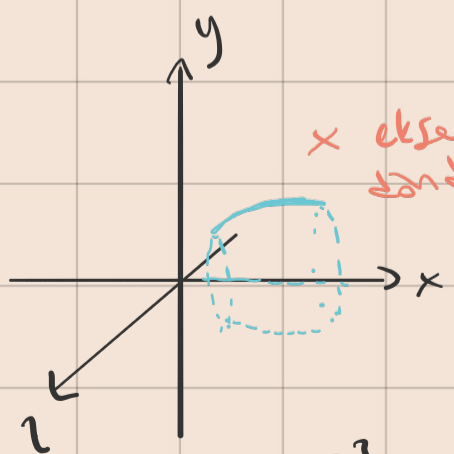
Örn
$$\int_0^3 \cos^2 x\,dx = \int \frac{1+\cos 2x}{2}dx$$

Örn
$$\int_0^{\frac{\pi}{2}} \sin x\cdot\cos^2 x\,dx = -\int_1^0 u^2\,du = \frac{u^3}{3}\Big|_1^0 = 0-\frac{1}{3} = \boxed{-\frac{1}{3}}$$

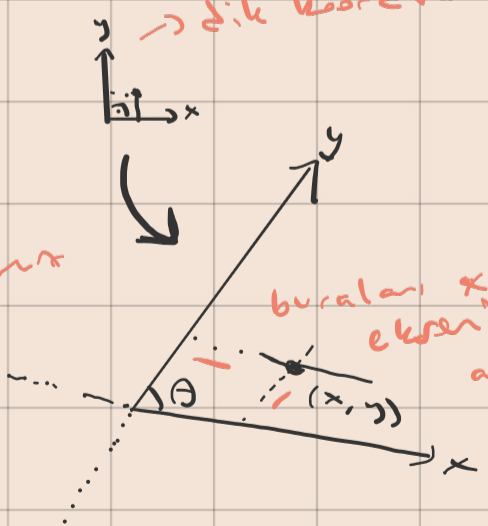
$u = \cos x$

$du = -\sin x\,dx$

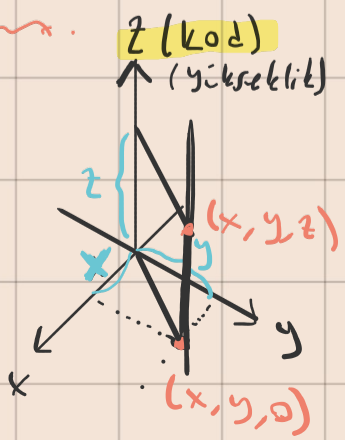
Dönel Cisim



x ekseninde döndürüyoruz



bucakları x ve y eksenine göre paralel alıyoruz.

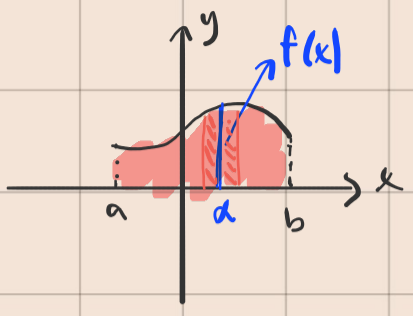


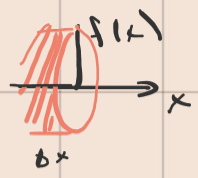
$$\mathbb{R}^3 = \mathbb{R}\times\mathbb{R}\times\mathbb{R} = \{(x,y,z) : x,y,z\in\mathbb{R}\}$$

uzay

Dönel Cismin Hacmi

1- Disk Metodu





Silindir: Bir doğrunun, bir eğri boyunca bir doğruya/doğrultuya paralel kalacak hareket etmesiyle oluşturduğu yüzeye silindir yüzeyi denir. $V = \pi \cdot r^2 \cdot h$



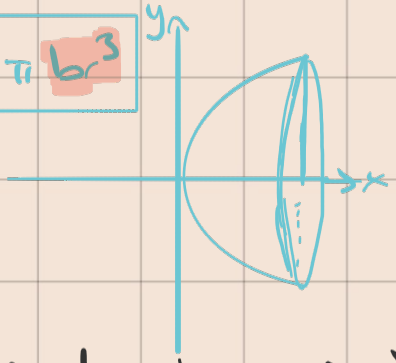
Bu dikdörtgenlerin etrafında döndürülmesi ile oluşan silindirlerin hacimlerinin toplamı, dönel cismin hacmini verir.

$$V_i = \pi \cdot \{f(x)\}^2 \cdot \Delta x \rightarrow V = \int_a^b \pi \cdot [f(x)]^2 \cdot dx = \pi \int_a^b y^2 \, dx$$

Örn

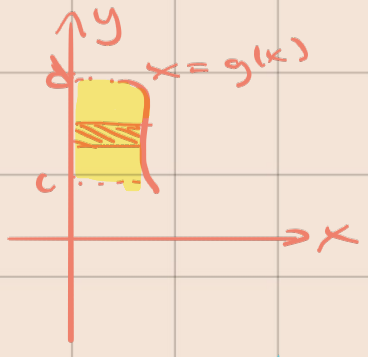
$y = 2\sqrt{x}$ eğrisinin $[0, 3]$ aralığında kalan kısmı ile x ekseni arasında kalan bölgenin x ekseni etrafında dönmesiyle oluşan cismin hacmi?

$$V = \pi \int_a^b y^2 \, dx = \pi \int_0^3 (2\sqrt{x})^2 = \pi \int_0^3 4x \, dx = 18\pi \text{ br}^3$$



Örn

$y = x^3$ eğrisinin $y = 1$ ve $y = 3$ doğruları arasında kalan bölgenin y ekseni etrafında dönmesi ile oluşan cismin hacmini bulunuz.



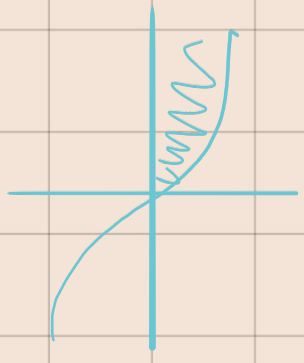
$$V = \pi \int_c^d x^2 \, dy = \pi \int_c^d [g(y)]^2 \, dy$$

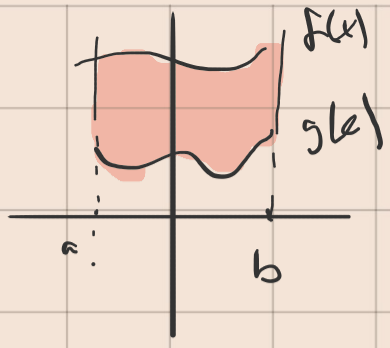
$y = x^3$

$x = \sqrt[3]{y}$

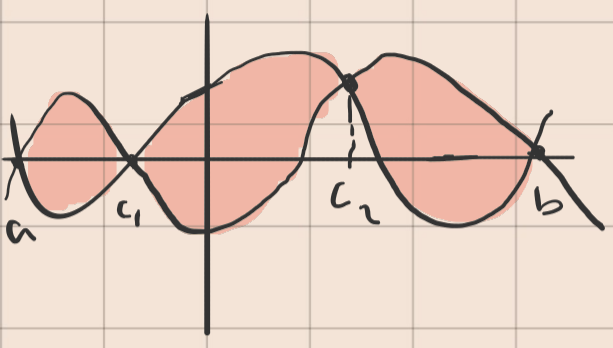
$$V = \pi \int_1^3 x^2 \, dy = \pi \int_1^3 (\sqrt[3]{y})^2 \, dy = \pi \cdot \left(\frac{3}{5} y^{\frac{5}{3}} \Big|_1^3 \right)$$

($\sqrt[3]{y}$ → $y^{\frac{1}{3}}$)





$$V = \pi \int_a^b \left([f(x)]^2 - [g(x)]^2 \right) dx$$



$$V = \pi \cdot \left(\int_a^{c_1} \left[f(x) \right]^2 - \left[g(x) \right]^2 + \int_{c_1}^{c_2} \cdots + \int_{c_2}^{b} \cdots \right)$$

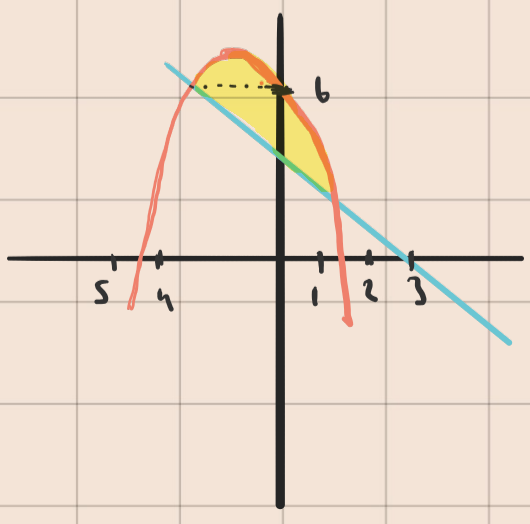
Örn $y = 2\sqrt{x}$

$x = 4$ doğrusu etrafında döndürülüyor.

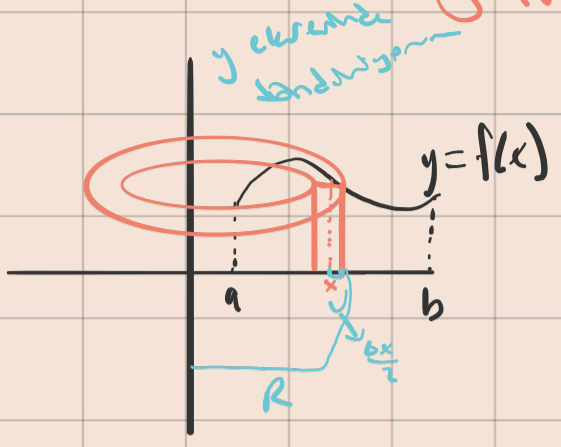


$$\pi \int_0^4 (4-x)^2 dy = \pi \int_0^4 \left(4 - \frac{y^2}{4}\right)^2 dy = \boxed{\frac{512}{15} \pi \, br^3}$$

Örn $y = -x^2 - 3x + 6$ parabolü ile $x + y - 3 = 0$ doğrusunun sınırladığı alan $x + y - 3 = 0$ doğrusu etrafında döndürülüyor.



2- Silindir Kabuk (soğan gibi)



$$V_1 = \pi \cdot R^2 \cdot f(x) - \pi r^2 \cdot f(x)$$

$$= \pi \cdot f(x) \cdot [R^2 - r^2]$$

$$= \pi f(x)(R-r)(R+r)$$

$$= \pi f(x) \cdot \Delta x \cdot 2x$$

$$= 2\pi \cdot x \cdot f(x) \cdot dx$$

$$V = 2\pi \int_a^b x f(x)\, dx$$

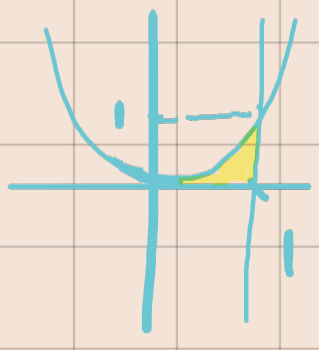
$$V_{oy} = 2\pi \int_a^b x \cdot y \cdot dx$$

$$R = x + \frac{\Delta x}{2}$$

$$r = x - \frac{\Delta x}{2}$$

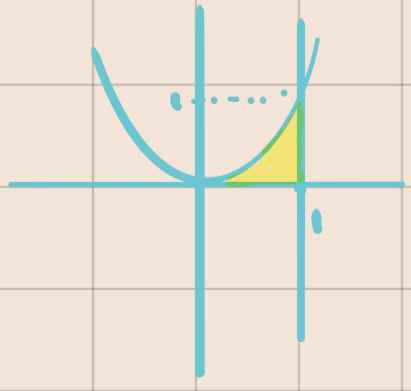
Örn

i-) $y = x^2$, $x=0$, $x=1$ ve x-ek. ile sınırlı bölge x-ekseninde döndürüldüğünde meydana gelen cismin hacmini bulunuz.



$$V_1 = \pi \int_0^1 y^2\, dx \quad , \quad V_2 = 2\pi \int_0^1 x \cdot y \cdot dx$$

ii-) $y = x^2$, $x=0$, $x=1$, x-ek. ile sınırlı bölge y-ek. etrafında döndürüldüğünde meydana gelen cismin hacmini bulunuz.



$$\int_0^1$$

Örn

i) $y = 2x - x^2$, x-ek. ile sınırlı bölge y-ek. dönerse

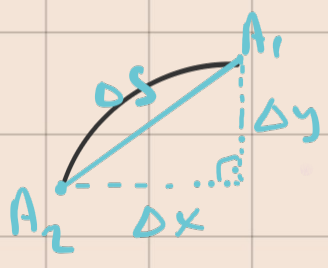
$$V = 2\pi \cdot \int_0^2 x \cdot y \cdot dx = 2\pi \int_0^2 x(2x - x^2)\,dx$$

ii) x-ek. dönderilirse?

$$V = \pi \cdot \int_0^2 y^2\,dx = \pi \int_0^2 (2x - x^2)^2\,dx$$

Yay Uzunluğu





$$\sqrt{(\Delta x)^2 + (\Delta y)^2} = \Delta s$$

$$\Delta x = \frac{b-a}{n}$$

$$A_0A_1 + A_1A_2 + \cdots + A_{n-1}A_n = \sum_{k=1}^{n} |A_{k-1}A_k|$$

$n \to \infty$ limitine, $y = f(x)$ $[a,b]$ aralığındaki yayının uzunluğu denir.

$$\sqrt{(\Delta x)^2 \left[1 + \left(\frac{\Delta y}{\Delta x}\right)^2\right]} = \Delta s$$

$$\frac{\Delta s}{\Delta x} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}$$

$$\lim_{\Delta x \to 0} \frac{\Delta s}{\Delta x} = \lim_{\Delta x \to 0} \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} = \sqrt{1 + \lim_{\Delta x \to 0} \left(\frac{\Delta y}{\Delta x}\right)^2} =$$

$$\sqrt{1 + (y')^2} = \frac{ds}{dx}$$

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = f'(x) = y' = \frac{dy}{dx}$$

$$\boxed{ds = \sqrt{1 + (y')^2}\,dx}$$ olur. ds'ye yay uzunluğu diferansiyeli denir

Diferansiyel

$y = x^2$, $dy = 2x\,dx$

$= f'(x)\,dx$

$$ds = \sqrt{1 + (y')^2}\,dx$$

$$\boxed{s = \int_a^b \sqrt{1 + (y')^2}\,dx}$$

Örn $y = x^2$

i. Diferansiyeli
ii. Yay uzunluğu diferansiyeli bulunuz.
iii. $[0,1]$ arası yay uzunluğu

1/ $\boxed{dy = 2x\,dx}$

2/ $ds = \sqrt{1+(y')^2}\,dx$

$= \sqrt{1+(2x)^2} = \boxed{\sqrt{1+4x^2}}$

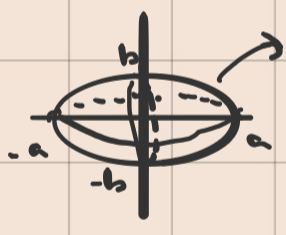
3/ $\int_0^1 \sqrt{1+4x^2}\,dx = \frac{1}{2}\int_0^1 \sqrt{1+u^2}\,du$

$2x = u$
$2\,dx = du$
$u = 1 \text{ ve } 1$

Ödev

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ Elips

$a^2 > b^2$, grafiğini bulunuz



elipsoid

Angel

Bir Fonksiyonun Ortalama Değeri

$y_{ort} = \frac{1}{b-a}\int_a^b f(x)\,dx$

Örn $y = 3x^2$, $[0,2]$ ort. değeri

$y_{ort} = \frac{1}{2-0}\int_0^2 3x^2\,dx$

$\int_0^1 \frac{dx}{\sqrt{1-x^2}} = \arcsin x\Big|_0^1$

Hes olmayan İntegraller

$y = f(x)$, $[a,b]$

$\int_0^1 \frac{dx}{\sqrt{1-x^2}}$, $x=1$ 'de sınırsızdır

$\int_0^{\infty} \frac{dx}{1+x^2}$, $\int_{-\infty}^{+\infty} \frac{dx}{1+x^2}$

1. Birinci tür hes olmayan int.

$\int_a^{\infty} f(x)dx$, $\int_{-\infty}^{b} f(x)dx$, $\int_{-\infty}^{+\infty} f(x)dx$

$$\int_a^{\infty} f(x)dx = \lim_{b\to\infty} \int_a^b f(x)dx$$

$$\int_{-\infty}^{b} f(x)dx = \lim_{a\to-\infty} \int_a^b f(x)dx$$

$$\int_a^b f(x)dx = \int_c^b f(x)dx + \int_a^c f(x)dx$$

$$\int_{-\infty}^{+\infty} f(x)dx = \int_{-\infty}^{c} f(x)dx + \int_c^{+\infty} f(x)dx = \lim_{a\to-\infty} \int_a^c f(x)dx + \lim_{b\to+\infty} \int_c^b f(x)dx$$

Örn

$$\int_0^{+\infty} \frac{dx}{1+x^2} = \lim_{b\to+\infty} \int_0^b \frac{dx}{1+x^2} = \lim_{b\to+\infty} (\arctan x|_0^b) = \lim_{b\to+\infty} (\arctan b - \arctan 0)$$

$\arctan 0 = u \Rightarrow \tan u = 0$ $(u=0)$

$\arctan b = v \Rightarrow \tan v = b$

Örn

$$\int_{-\infty}^{0} e^x\,dx = \lim_{a\to-\infty}\int_a^0 e^x\,dx = \lim_{a\to-\infty}\left(e^x\Big|_a^0\right) = \lim_{a\to-\infty}\left(e^0-e^a\right) = \lim_{a\to-\infty}(1-e^a) = 1$$

Örn

$$\int_{-\infty}^{+\infty}\frac{e^x}{1+e^{2x}}\,dx = \int_0^{+\infty}\frac{e^x}{1+e^{2x}}\,dx + \int_{-\infty}^{0}\frac{e^x}{1+e^{2x}}\,dx =$$

$e^x = u$

$e^x dx = du$

$$\lim_{b\to+\infty}\int_0^b \frac{e^x}{1+e^{2x}}\,dx + \lim_{a\to-\infty}\int_a^0 \frac{e^x}{1+e^{2x}}\,dx = \lim_{a\to-\infty}\left(\arctan e^x\Big|_a^0\right) + \lim_{a\to+\infty}\left(\arctan e^x\Big|_0^b\right)$$

$$= \lim_{a\to-\infty} e^a = 0$$

★ $\infty-\infty$ belirsizdir. Ancak $\infty-\infty$ sıfır ise o integral yakın. Yani integral yakınsaktır.

Örn

$$\lim_{k\to+\infty}\int_{-k}^{k} x\,dx = \lim_{k\to+\infty}\left(\frac{x^2}{2}\Big|_{-k}^{k}\right) = 0$$

★ integral $=1$ ise yakınsaktır.

Örn

$$\int_0^1 \ln x\,dx = \lim_{a\to0^+}\int_a^1 \ln x\,dx = \lim_{a\to0^+}\left(x\ln x - x\right)\Big|_a^1 = \lim_{a\to0^+}\left[(0-1) - a\ln a + a\right]$$

$x\ln x - x + c$

$[0,1]$ kapalı: için $\frac{c}{0}$ yakınsak

$0.\infty$ belirsizliği

$\lim_{a\to0^+}(a\ln a) = 0$ (l'Hopital ile)

$$= \boxed{-1}$$

Ödev

$$\int_0^{+\infty} \frac{dx}{\sqrt{x}\cdot(x+1)} = \int_0^{1} \frac{dx}{\sqrt{x}\cdot(x+1)} + \int_1^{+\infty} \frac{dx}{\sqrt{x}\cdot(x+1)} = \lim_{a\to 0^+} \int_a^1 \frac{dx}{\sqrt{x}\cdot(x+1)} + \lim_{b\to+\infty} \int_1^b \frac{dx}{\sqrt{x}\cdot(x+1)}$$

$\int \frac{2u\,du}{u(u^2+1)} = 2\int \frac{du}{u^2+1}$

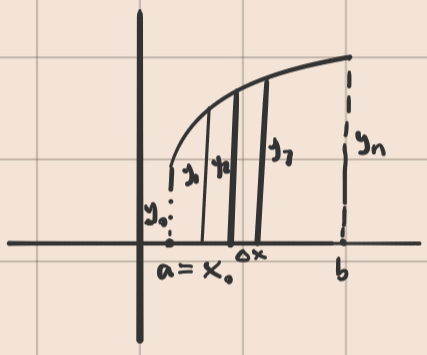
$x = u^2$
$dx = 2u\,du$

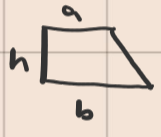
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Yaklaşık İntegrasyon

1) Yamuklar Kuralı

* Alan hesabında grafiğin altında kalan alanı almamışsın bu arada





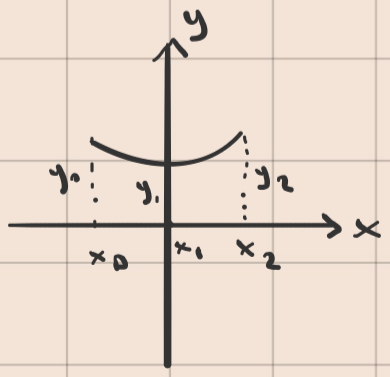
$A = \frac{a+b}{2}\cdot h$



$$h = \frac{b-a}{n} = \Delta x$$

$$A = \frac{1}{2}h\cdot(y_0+y_1) + \frac{1}{2}h\cdot(y_1+y_2) + \cdots + \frac{1}{2}\cdot h\cdot(y_{n-1}+y_n) = h\left(\frac{1}{2}y_0 + y_1 + y_2 + \cdots + y_{n-1} + \frac{1}{2}y_n\right)$$

2) Simpson Metodu



$y = ax^2 + bx + c$

$$\int_{-h}^{h} (ax^2+bx+c)\,dx = a\frac{x^3}{3} + b\frac{x^2}{2} + cx \Big|_{-h}^{h}$$

$$= a\frac{h^3}{3} + b\frac{h^2}{2} + ch + a\frac{h^3}{3} - a\frac{h^2}{2} + ch$$

$$= 2a\frac{h^3}{3} + 2ch = \frac{h}{3}(2ah^2 + bc) \text{ olur.}$$

$y = ax^2+bx+c$ parabol geçsin

$x = h$ için, $y = ah^2 + bh + c = y_0$

$x = 0$ " , $y = c = y_1$

$x = -h$ " , $y = ah^2 - bh + c = y_2$

$\Big\rangle$ $y_0 + y_1 + y_2 = 2ah^2 + bc$ bulunur.

$A = \frac{h}{3}(y_0 + 4y_1 + y_2)$ $y = f(x)$, $[a,b]$ aralığındaki yayını bu şekilde parçalı parçalara benzetelim. $h = \frac{b-a}{n}$, n çift

Örnek

$$-\int_0^{\frac{1}{2}} [[2-x]]\,dx = \int_0^{\frac{1}{2}} 1\,dx = \boxed{\frac{1}{2}}$$

$0 < x < \frac{1}{2}$ için $[[2-x]] = 1$

$$-\int_{-1}^{2} \frac{x^h}{|x|}\,dx = \int_{-1}^{0} \frac{x^h}{|x|}\,dx + \int_0^2 \frac{x^h}{|x|}\,dx$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



$$-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot x \cdot \ln(\sin x)\,dx =$$

$\ln(\sin x) = u$

$[\ln(\sin x)]' = \frac{\cos x}{\sin x}$

$(\ln u)' = \frac{u'}{u}$

Çok Değişkenli Fonksiyonlar — Bir acto boyut

$f: D \to \mathbb{R}$

- $(x,y) \to F(x,y) = z$ iki değişkenli fonk. x ve y bağımsız değişkenler, z bağımlı değişken

$z = x + y$, $z = x^2 + y^2$, $z^2 = x^2 + y^2$, $z = \cos(x+y)$

- Silindirin hacmi $V = \pi \cdot r^2 \cdot h$

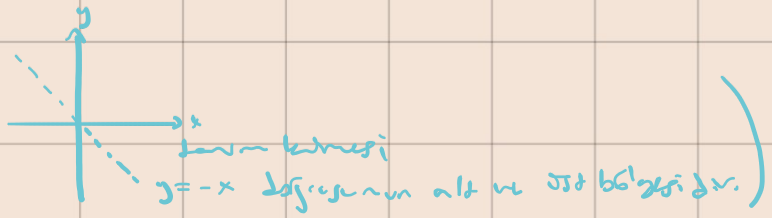
$r = x$, $h = y$, $z = \pi \cdot x^2 \cdot h$

Tanım bölgesi: $z = F(x,y)$, x ve y'nin alacağı değerler bölgesi fonksiyonun tanım bölgesidir.

Örn

i) $z = x + y$, $R \times R = R^2$

ii) $z = \frac{1}{x+y}$, $x + y \neq 0$ ($x+y=0$ olamıyo ya $y = -x$)



tanım kümesi

Örn

i) $f(x,y)=z$, (a,b) nok. tanımlı.

ii) $\lim_{(x,y)\to(a,b)} f(x,y)=L$

iii) $L=f(a,b)$ ise $z=f(x,y)$ fonk. (a,b) nok. süreklidir.

$$\lim_{\substack{x\to a\\ y\to b}} f(x,y) = \lim_{(x,y)\to(a,b)} f(x,y)$$

$$f(x,y)=\begin{cases} 0, & (x,y)=(0,0) \\ \frac{2xy}{x^2+y^2}, & (x,y)\neq(0,0) \end{cases}$$

$(0,0)$ nok. sürekli midir?

$\lim_{(x,y)\to(0,0)} \left(\frac{2xy}{x^2+y^2}\right)$

$= \lim_{y\to 0} \frac{2\cdot 0\cdot y}{0+y^2} = 0$

$\lim_{\substack{x\to 0\\ y\to 0}} \frac{2xy}{x^2+y^2} = \lim_{x\to 0} \frac{2x\cdot 0}{x^2+0} = 0$

$y=k\cdot x$ doğruları boyunca $(0,0)$ nok. yaklaşabiliyor.

$\lim_{(x,y)\to(0,0)} \frac{2xy}{x^2+y^2} = \lim_{x\to 0} \left[\frac{2x\cdot kx}{x^2+(kx)^2}\right]$

$= \lim_{x\to 0} \frac{2kx^2}{x^2(1+k^2)}$

$= \frac{2k}{1+k^2}$ olup limit k değerine bağlıdır. Limit yoktur.

Not: $y=kx^2$ boyunca $(0,0)$ nok. yaklaşarak limiti hesaplayınız.

Örnekler

$\lim_{(x,y)\to(0,0)} \frac{e^y\cdot \sin x}{x}$

Kısmi Türev

$$\lim_{\Delta x\to 0} \frac{f(x+\Delta x,y)-f(x,y)}{\Delta x} = \frac{\partial f}{\partial x} = z'_x = z_x = f_x = \frac{\partial z}{\partial x} = f_x(x,y)$$

$$\lim_{\Delta y\to 0} \frac{f(x,y+\Delta y)-f(x,y)}{\Delta y} = \frac{\partial f}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y(x,y)$$

Örn

i. $z=x^2+y^2$

$z_x=\frac{\partial z}{\partial x}=2x$, $z_y=\frac{\partial z}{\partial y}=2y$

ii. $z=x^2y+3xy^4$, $z_x=2xy+3y^4$

$z_y=x^2+12xy^3$

Örn

$ye^{x^2}+x\ln(y^2-x^2)=z$

$z_x=\frac{\partial z}{\partial x}=2xye^{x^2}+\ln(y^2-x^2)+x\cdot\frac{-2x}{y^2-x^2}$

$z_y=\frac{\partial z}{\partial y}=e^{x^2}+x\cdot\frac{2y}{y^2-x^2}$

Örn

$f_x(4,-5)$, $f_y(4,-5)$

Örn

$f(x,y) = y \cdot \sin(x \cdot y)$

$\frac{\partial f}{\partial x} = y \cdot y \cos(x \cdot y)$

$\frac{\partial f}{\partial y} = \sin(xy) + y \cdot x \cdot \cos(xy)$!!!

Yüksek Mertebeden Kısmi Türevler

$z = f(x,y)$ fonk. z_x, z_y kısmi türevleri yine iki değişkenli fonk.

$\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2} = z_{xx}$

$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2} = z_{yy}$

$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y \partial x} = z_{xy}$

Örn

$z = \cos(x+y)$

$\frac{\partial z}{\partial x} = -\sin(x+y)$

Örn

$f(x,y) = 2x^2 - 3y - 4$

$f(x,y) = \sqrt{x^2+y^2}$ $\quad \frac{\partial f}{\partial x} =$

Bileşke fonksiyonda

$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$ yani $\quad \frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$

$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$

Örn

$z = e^x \sin y, \quad x = \ln t, \quad y = t^2$

$\frac{dz}{dt} = \left(e^{\ln t} \cdot \sin t^2\right)' = \left(t \cdot \sin t^2\right)' \cdot \,' = \frac{d}{dt}$

$\frac{\partial z}{\partial x} = e^x \cdot \sin y, \quad \frac{\partial z}{\partial y} = e^x \cos y, \quad \frac{dx}{dt} = \frac{1}{t}, \quad \frac{dy}{dt} = 2t$

$$z = \ln(x^2+y^2)$$

$$dz = \frac{2x}{x^2+y^2}\,dx + \frac{2y}{x^2+y^2}\cdot dy$$

Kısmi Türevde Kapalı Fonksiyon Türevi

$x^2 \cdot y + z^3 x + zy = x^2 + y^2 - 1$ ise $\frac{\partial z}{\partial x}$ nedir?

y sabit sayı

z → x'e bağlı fonksiyon

x

$$2xy + 3z^2 \cdot \frac{\partial z}{\partial x} \cdot x + z^3 \cdot 1 + \frac{\partial z}{\partial x} y = 2x$$

ardından $\frac{\partial z}{\partial x}$'i yalnız bırakınca türevi bulmuş olunur

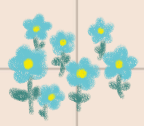
$$\frac{\partial z}{\partial x}(3z^2x + y) = 2x - 2xy - z^3$$





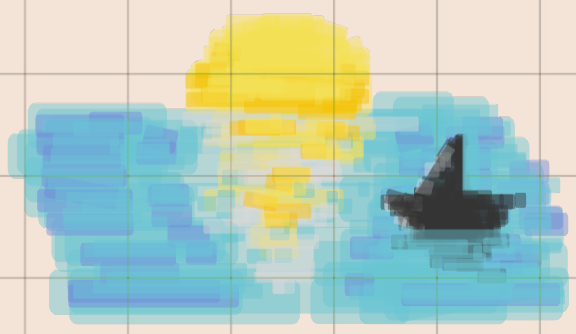
She does not GAF.











Yerel Ekstremum Noktalarını Bulmak

$$D = f_{xx} \cdot f_{yy} - (f_{xy})^2$$

1) $D(x_0,y_0) > 0$, $f_{xx}(x_0,y_0) > 0$ yerel min
2) $D(x_0,y_0) > 0$, $f_{xx}(x_0,y_0) < 0$ yerel max
3) $D(x_0,y_0) < 0$ ise (x_0,y_0) Eyer noktası
4) $D(x_0,y_0) = 0$ ise BAŞKA Yollar

Örn $f(x,y) = x^2 + 2xy$ yerel max, min, eyer noktaları?

$f(x) = 0 \rightarrow 2x + 2y = 0$ ($y=0$) $\rightarrow 2x = 0 \rightarrow x = 0$ (0,0) kritik nokta

$f(y) = 0 \rightarrow 2x = 0 \rightarrow x = 0$

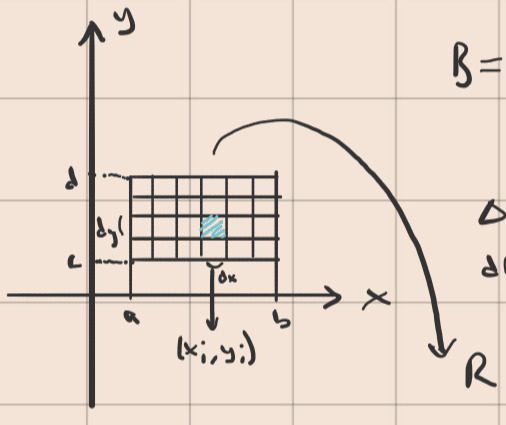
$f_{xx} = 2 \quad f_{xy} = 2 \quad f_{yy} = 0$

f_{xy}: x'e göre aldığımız y'ye göre diyor

$D = 2 \cdot 0 - 4 = \boxed{-4}$

(0,0) eyer noktasıdır.

İki Katlı İntegral



$$B = \{(x,y) : a \le x \le b, \; c \le y \le d\} \qquad f: B \rightarrow R, \quad (x,y) \rightarrow z = f(x,y)$$

$$\Delta A = \Delta x \cdot \Delta y = \Delta y \cdot \Delta x$$

$$dA = dx \cdot dy = dy\,dx$$

$$\lim_{n \to \infty} \sum f(x_i, y_i) \cdot \Delta A_i = \iint_B f(x,y)\,dA$$

$$= \iint_B f(x,y)\,dy\,dx = \iint_B f(x,y)\,dx\,dy$$

$$= \int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_a^b f(x,y)\,dx\,dy$$

$$= \int_a^b \left[\int_c^d f(x,y)\,dy \right] dx = \int_c^d \left[\int_a^b f(x,y)\,dx \right] dy$$

Not: $f(x,y) = 1$ ise,

$$\iint_B 1 \cdot dy\,dx = \int_a^b \left[\int_c^d dy \right] dx = \int_a^b (d-c)\,dx = (d-c) \int_a^b dx = (d-c)(b-a)$$

Örn

$$\int_0^2 \left[\int_0^1 (4-y-x)\,dy \right] dx = \int_0^2 \left(4y - \frac{y^2}{2} - xy \Big|_0^1 \right) dx$$

Örn

$$\int_1^2 \left(\int_0^4 2xy\,dy \right) dx = \int_1^2 xy^2 \Big|_0^4\,dx = \int_1^2 16x\,dx = 8x^2 \Big|_1^2 = 4 \cdot 8 - 8 = \boxed{24}$$

$$\int \frac{du}{1+u} = \ln|1+u| + c$$

Örn

$$\iint_R xy\cos y \, dA, \quad R: -1 \le x \le 1, \ 0 \le y \le \pi$$

$$= \int_0^{\pi} \left[\int_{-1}^{1} xy\cos y \, dx \right] dy = \int_0^{\pi} \frac{x^2 y \cos y}{2} \Big|_{-1}^{1} dy = \int_0^{\pi} y\cos y \, dy =$$

(k. sm integrasyon gerekir)

Örn

$$\int_0^1 \int_y^1 \frac{\sin x}{x} \, dx \, dy =$$

Örn

$$\int_0^2 \int_{x^2}^{2x} (4x+2) \, dy \, dx = \int_0^4 \int_{y/2}^{\sqrt{y}} (4x+2) \, dx \, dy = \int_0^2 (4x+2) y \Big|_{x^2}^{2x} dx = \int_0^2 (4x+2)(2x-x^2) dx$$

Örn

$$\int_0^1 \int_x^1 e^{y^2} \, dy \, dx \xrightarrow{\text{hesaplanamadığı için sıra değiştirilir}} = \int_0^1 \int_0^y e^{y^2} \, dx \, dy = \int_0^1 e^{y^2} \cdot x \Big|_0^y dy$$

$$= \int_0^1 e^{y^2} \cdot y \, dy$$

ilk / çözüt kolay integrallerde değişken değiştirme